Notes 22,23 and 24

☁️ Azure Interaction & Automation – Easy Notes

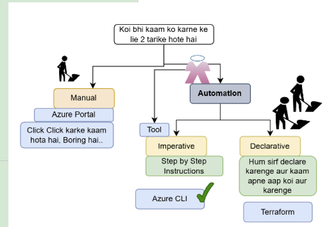
1. Azure ke saath kaise interact karte hain?

Azure resources ko manage karne ke 2 main tareeke hote hain:

🖱️ (A) Manual Method (Azure Portal – GUI)

- Browser based hota hai (Azure Portal)
- “Click-click” karke kaam hota hai
- Easy hai, but **slow & non-automated**
- Large scale projects ke liye useful nahi

👉 Isko generally “boring / manual work” bola jata hai kyunki automation nahi hoti

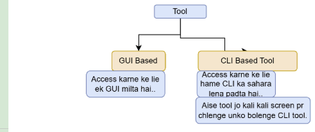


💻 (B) Tool Method (CLI – Command Line Interface)

- Black screen (PowerShell / Terminal) use hota hai
- Commands likh ke Azure control karte hain
- Fast + automation friendly
- Real DevOps yahin hota hai

👉 Example tools:

- Azure CLI (`az`)
- AWS CLI (`aws`)
- GCP CLI (`gcloud`)



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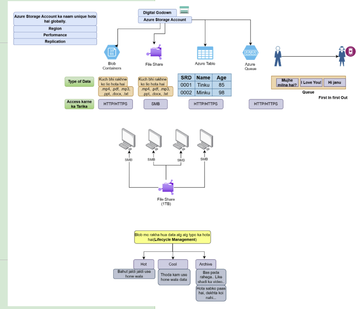
2. Infrastructure Management Styles

Azure/DevOps mein 2 thinking styles hote hain:



⚙️ (A) Imperative Approach (Step-by-Step)

👉 “Kaise karna hai” batana padta hai

- Tum system ko exact steps dete ho
- Har action manually define hota hai
- Example: “pehle VM banao, phir network attach karo”

🔧 Tool:

- **Azure CLI**

👉 Simple meaning:

“Main instructions deta hoon, system follow karta hai”

🎯 (B) Declarative Approach (Goal-Oriented)

👉 “Kya chahiye” bas ye batana hota hai

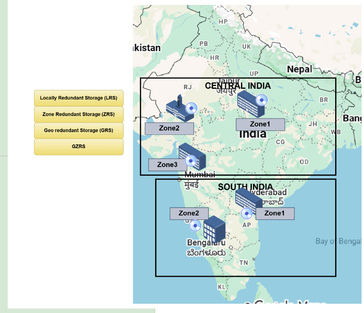
- Tum final state define karte ho
- System khud steps decide karta hai
- Example: “Mujhe 2 VM + network chahiye”

🔧 Tool:

- **Terraform**

👉 Simple meaning:

“Result batao, kaise banana hai system samjhega”



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3. Azure CLI Basics

Azure CLI ek tarah ka **wrapper hai Azure REST API ka**

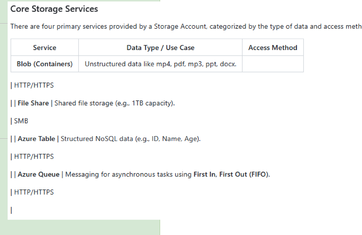
👉 Simple meaning:

- Direct API call karna tough hota hai
- CLI usko easy commands mein convert karta hai

Example idea:

```
/vm/create → az vm create
```

Azure Resource Manager (ARM) ke through backend mein sab hota hai.



Core Storage Services

There are four primary services provided by a Storage Account, categorized by the type of data and access method:

Service	Data Type / Use Case	Access Method
Blob (Containers)	Unstructured data like mp4, pdf, mp3, ppt, docx.	

| HTTP/HTTPS

| | **File Share** | Shared file storage (e.g., 1TB capacity).

| SMB

| | **Azure Table** | Structured NoSQL data (e.g., ID, Name, Age).

| HTTP/HTTPS

| | **Azure Queue** | Messaging for asynchronous tasks using **First In, First Out (FIFO)**.

| HTTP/HTTPS

|

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4. Installation & Setup

📥 Install Azure CLI

- `.msi` file download karo (Microsoft website)
- Next → Next → Install

✅ Verify Installation

PowerShell/Terminal mein likho:

```
az
```

Agar help menu aa gaya → installation successful ✔️

🔐 Login (Authentication)

```
az login
```

- Browser open hoga
- Microsoft account se login karna hota hai

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5. Important Basic Commands

📚 Help command

```
az --help
```

👉 Kisi bhi command ka usage samajhne ke liye

📌 Subscription commands

Task	Command
Subscriptions list karna	`az account list`
Active subscription set karna	`az account set`

📁 Resource Group commands

Task	Command
Resource group create karna	`az group create`
Resource groups dekhna	`az group list`
Specific group details	`az group show`

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6. Automation & DevOps Goal 🚀

Final aim hota hai:

🔄 Step 1: Code likhna

- Infrastructure ko code ke form mein define karna

🔄 Step 2: CI/CD pipeline

- Automatic build + deploy process

🔄 Step 3: Branching Strategy

- Code management (Git branches etc.)

👉 Matlab:

Manual work hatao → automation build karo → system self-managed banao

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💡 Final Motivation Line (Simple Hinglish)

“Jab insaan mehnat aur discipline lagata hai, toh impossible cheezein bhi possible ho jaati hain.”

Agar tu chahe toh main isko:

- 🧠 One-page cheat sheet
- 📘 Exam questios + answers
- 🚀 Real Azure CLI practice lab
  mein bhi convert kar sakta hoon.

1. Azure Hierarchy (Structure of Azure)

Azure ko ek **bank system** ki tarah samajh lo, jisme levels hote hain:

🏦 (A) Entra ID & Management Groups

- Top-most level hota hai
- Users, identity aur access control yahin se manage hota hai
- Multiple subscriptions ko group kar sakte ho

👉 Simple meaning:

"Kaun kya access karega" yahan decide hota hai

💳 (B) Subscriptions

- Ye billing unit hota hai
- Har subscription ka apna budget hota hai (e.g. ₹18,000)
- Resources ka cost yahan track hota hai

👉 Simple meaning:

"Paise aur usage ka account"

📦 (C) Resource Groups (RG)

- Logical container hota hai
- Related resources ek jagah group hote hain
- Example: VM + Storage + Network ek RG mein

👉 Banane ke 2 tareeke:

- Portal (Click-click method 🖱️)
- Azure CLI (`az` commands 💻)

🌍 Important Rule

⚠️ Storage Account Name Must Be Globally Unique

- Matlab duniya mein kisi aur ka same name nahi ho sakta
- Example:
  - `mystorage123` ✔️ (unique)
  - same name kisi aur ne liya → ❌ not allowed

Azure Storage Account (Digital Godown 🏪)

Storage Account ko samjho:

"Ek digital warehouse jahan sab data store hota hai"

📂 Storage Services (4 Types)

Service	Kya store hota hai	Kaise access hota hai
🟦 Blob Storage I	mages, videos, PDFs, MP3, docs (unstructured data)	HTTP/HTTPS
📁 File Share	Shared folder system (like network drive)	SMB protocol
🗃️ Table Storage	Simple NoSQL data (ID, Name, Age type data)	HTTP/HTTPS
📩 Queue Storage	Messages for communication between apps (FIFO)	HTTP/HTTPS

🧠 Easy Understanding:

Blob → media files 📸🎥

File Share → shared drive 🗂️

- Table → simple database 📊
- Queue → messaging system 📬

3. Data Lifecycle Management (Blob Storage Tiers)

Azure cost save karne ke liye data ko different storage levels mein move karta hai:

🔥 Hot Tier

- Frequently accessed data
- Example: daily use files, active projects

👉 Fast access, but costly

❄️ Cool Tier

- Rarely accessed data
- Example: old project files

👉 Balanced cost + access

🧊 Archive Tier

- Almost never used data
- Example: old wedding videos 😂

👉 Cheapest storage, but slow access

🧠 Simple idea:

"Jitna kam use, utna sasta storage"

4. Azure Regions, Zones & Data Safety

Azure duniya bhar mein data centers use karta hai.

🌍 Regions

- Geographic locations (e.g. West US, East US)
- Har region ek bada area hota hai

🏢 Availability Zones

- Ek region ke andar multiple physical data centers
- Agar ek fail ho jaye, doosra chal jata hai

5. Data Redundancy (Backup System 🔁)

Azure data ko multiple copies mein store karta hai for safety:

🟢 (A) LRS – Locally Redundant Storage

- Same data center ke andar copies
- Basic protection

👉 Simple:

"Ek hi building mein backup"

🔵 (B) ZRS – Zone Redundant Storage

- Same region ke different zones mein copies
- Better protection

👉 Simple:

"Ek city ke different buildings mein backup"

🟣 (C) GRS – Geo Redundant Storage

- Different regions (e.g. India + US)
- Highest level of safety

👉 Simple:

"Different countries mein backup"

🧠 Quick Summary:

Type	Backup Location	Safety Level
LRS	Same data center	Low
ZRS	Same region, different zones	Medium
GRS	Different regions	High

🚀 Final Simple Summary

- Azure structure = Bank system jaisa hierarchy
- Subscriptions = billing account
- Resource Groups = container for resources
- Storage Account = digital warehouse
- Blob/File/Table/Queue = different storage types
- Hot/Cool/Archive = cost-based storage levels
- LRS/ZRS/GRS = data protection levels

💡 One-Line Revision

Azure = Identity (Entra ID) → Billing (Subscription) → Containers (RG) → Storage (Data services) → Safety (Redundancy)